

SEDAR 32: South Atlantic Blueline Tilefish and Gray Triggerfish

Blueline Tilefish

Data Workshop

Life History

- Stock Structure
 - Blueline tilefish stock definition needs to be investigated further. Genetic study or some other form of stock identification study needs to be undertaken with samples (muscle, fin clips, etc.) collected from several locations within the Gulf of Mexico and the northwestern Atlantic.
 - Habitat studies of deep water sites in the mid-Atlantic, specifically Norfolk Canyon, Baltimore Canyon, and Hudson Canyon need to be undertaken. Temperature data from research conducted in the 1970s in Norfolk Canyon can be used for comparison purposes.
- Age Data
 - Age readings of blueline tilefish need to be validated. Within and between lab variability in readings is large and needs to be addressed. The potential bias in age readings between laboratories also needs to be addressed with another age workshop and exchange of calibration sets of samples.
 - Marginal increment analysis needs to be undertaken in order to convert increment counts to calendar ages. Samples processed and read in older studies will need to be re-examined and margin codes recorded for each.
 - More recreational fishery age samples need to be collected.
- Reproductive Biology Data
 - Overall, more reproductive samples need to be collected. Because small, young fish were lacking from the biological collections, specimens under 18 inches will be needed to address age and size at maturity. Whole gonads will need to be collected for a fecundity study. Specimens collected from throughout the species range and covering all months of the year are needed to better describe spawning season and spawning periodicity.
- Ad-hoc Discard Mortality Sub-group
 - Future research is needed to examine discard mortality rates for this species, as well as factors that affect survival (e.g., gear type, temperature, depth).

Commercial Statistics

- Discard
 - Investigate the validity and magnitude of “no discard” trips. This may include fisher interviews throughout the region.
 - Examine potential impacts on “no discard” trips, including:
 - Trip length

- Trip dates in relation to fishery regulations
 - Trip targeting
 - Trip area fished
- Improve discard logbook data collections via program expansion or more detailed reporting (e.g. more detailed logbook, electronic reporting)
- Develop an observer program that is representative of the fishery in the South Atlantic.
- Biosampling
 - Standardize TIP sampling protocol to get representative samples at the species level.
 - Develop an observer program that is representative of the fishery in the South Atlantic.
 - Increase untargeted sampling in NE and Mid-Atlantic observer programs.
 - Increase untargeted dockside sampling in NE and Mid-Atlantic.

Recreational Statistics

- Continued research efforts to incorporate/require logbook reporting from recreational anglers.
- Quantify historical fishing photos for use in future SEDARs.
- Fund research efforts to collect discard length and age data from the private sector.
- Improve metadata collection in the recreational fishery.
- Pre-stratify MRIP Keys, N-S Canaveral, N – S Hatteras.
- Research possibility of implementing private recreational reef fish stamp to determine universe and reporting strategies.
- At-sea observers collect surface and bottom temperature.
- At-sea observer protocols should include all fields currently used in FL i.e., condition and depth of released fish.

Indices

- Evaluate various sub-setting methods to identify effective effort. Methods that have been applied or considered include in this and previous SEDAR assessments include the Jaccard statistic, Stephens and MacCall approach, variations of Stephens and MacCall approach (e.g., using amount of catch rather than presence-absence), and other multivariate statistical approaches (e.g., cluster analysis).
- Evaluate various standardization methods to handle zeros in the catch, e.g., delta-GLM, zero-inflated Poisson, zero-inflated negative binomial, hurdle models, etc.
- Evaluate possible effects of circle hooks on catchability of reef fishes.
- Need fishery independent sampling of deep-water species, including blueline tilefish. Need funding to support these efforts.

Assessment Process

The assessment panel made the following recommendations.

- Develop a fishery independent sampling program for abundance of the deepwater snapper-grouper complex (including blueline tilefish). Fishery dependent abundance indices used in this assessment were uncertain in part due to the lack of an effective sampling methodology.
- Implement a systematic age sampling program and systematic evaluation of aging error. Age samples were important in this assessment but reasonable sample sizes were only available for the last 3-4 years of the assessment.
- Better characterize reproductive parameters including age at maturity, batch fecundity, spawning seasonality, and spawning frequency.
- Better characterize the genetic structure of the stock and evaluate the possibility of local population structure.
- Better characterize the inshore-offshore migratory dynamics of the stock and the degree of fidelity to spawning areas. Portions of the stock may be further offshore in some years and hence not available to the fishery.
- Age-dependent natural mortality was estimated by indirect methods for this assessment of blueline tilefish. Tag-recapture programs may prove useful for estimating mortality.

Review Workshop

Research recommendations for blueline tilefish were provided in the data and assessment working group documents. The Panel noted that many of these recommendations reflected concerns across a range of deep-water species and therefore confined their attention to those specific to the stock assessment of blueline tilefish.

While the panel supports work on stock structure, we recommend starting with the available information on describing the differences in demographics/life history characteristics over the range of the management area. Additionally, the available information on habitat in the areas listed should be evaluated before initiating any new studies.

Given that this is an age-based assessment, the comparison and calibration studies for the age determination should receive high priority along with the marginal increment analysis to determine if the opaque zone is formed annually. Many species would probably benefit from expanding the MRIP program to include age sampling.

The collection of information to better describe spawning season and spawning periodicity could probably start with fishery-dependent sources, but will need data from fishery-independent programs to cover the range of the species. The latter program would probably have to be tailored to provide samples across the deep-water snapper/grouper complex.

Studies of discard mortality should be low priority given the current negligible discard rate in the commercial fishery. The collection of additional information on discards and catch (e.g, lengths, ageing material) is important especially for the areas north of Hatteras, but would likely require an observer program developed for all fisheries focusing on the deep-water snapper/grouper complex.

The BAM model is reliant on historical information and any data on size compositions, maximum size, etc., that can be obtained from historical recreational fishing photos could be quite useful. One of the main issues raised about the recreational fishery concerned the high landings in the mid-late 2000s, especially the high landing and discard estimates for 2007. Closer scrutiny of these estimates requires data at higher resolution than was apparently available for this stock assessment.

With respect to developing a fishery-independent survey, sampling of deep-water habitats may elucidate habitat characteristics, and spatial distributions of blueline tilefish and other deep-water reef fishes. If a sufficient time series is developed, then a fishery-independent index may be developed.

Gray Triggerfish

Data Workshop

Life History

- Stock Structure
 - Gray triggerfish stock definition needs to be investigated further. Genetic study or some other form of stock identification study needs to be undertaken with samples (muscle, fin clips, etc.) collected from several locations within the Gulf of Mexico and the northwestern Atlantic. Special attention should be given to fish from the southern tip of Florida, to determine stock split for the South Atlantic and Gulf of Mexico.
- Age Data
 - Age readings of gray triggerfish need to be validated. Uncertainty in the interpretation of the zones formed on the spines of the triggerfish is high.
 - More recreational age samples are needed to better characterize the catch.
- Reproductive Biology Data
 - Overall, more reproductive samples need to be collected. Fecundity data for the South Atlantic was not available. Whole gonad samples will need to be collected to estimate fecundity of this fish. Samples are also needed from the entire region and all months of the year to better describe spawning season and spawning frequency.
- Ad-hoc Discard Mortality Sub-group
 - The majority of the discard mortality rates discussed for gray triggerfish are for fish observed at the surface. There is no information on delayed sources of mortality for discarded fish. Research investigating the factors that impact discard survival (i.e., gear, hook type, depth, presence and type of barotrauma) is needed.

Commercial Statistics

- Landings
 - Require species level reporting in state trip ticket programs.
- Discard

- Investigate the validity and magnitude of “no discard” trips. This may include fisher interviews throughout the region.
- Examine potential impacts on “no discard” trips, including:
 - Trip length
 - Trip dates in relation to fishery regulations
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